



Society of Petroleum Engineers

SPE-229553-MS

A Disruptive Approach for Rock Typing Prediction and Modeling in Carbonate Reservoirs

G rard J. Massonnat, TotalEnergies, Pau, France; Mickael Barbier and Lucie Dal Soglio, Akkodis, Pau, France; Charles Danquigny, TotalEnergies, Pau, France / Avignon Universit , France; Julien Michel, Akkodis, Pau, France; Jean-Louis Lesueur, TotalEnergies, Pau, France

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229553-MS](https://doi.org/10.2118/229553-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Carbonate reservoirs, despite their global significance in hydrocarbon production, present a persistent challenge to reservoir characterization due to their intrinsic heterogeneity. This complexity stems from a combination of depositional variability, diagenetic overprint, and the presence of diverse pore systems, which disrupt conventional petrophysical relationships. In this context, rock typing—the classification of reservoir rocks into groups with internally consistent flow and storage properties —emerges as a critical step in bridging geological description and dynamic reservoir modeling.

The necessity of rock typing in carbonate systems is not merely academic; it is vital. Unlike siliciclastic reservoirs, where depositional facies often correlate well with petrophysical behavior, carbonate rocks exhibit a wide spectrum of pore types and diagenetic alterations that hamper such correlations. Effective rock typing enables the grouping of rocks with similar petrophysical responses, thereby facilitating the assignment of porosity-permeability relationships, capillary pressure functions, and relative permeability curves within reservoir models (Lucia, 1995; Jennings & Lucia, 2003).

The central challenge involves transferring rock type information from core-scale observations to reservoir-scale models. Core data offer detailed insights into petrophysical properties, but their spatial coverage is limited, and extending these findings to three-dimensional reservoir grids can lead to significant bias and mistakes. Additionally, the spatial distribution of rock types is governed by geological processes that may not be well defined due to limited available data, particularly in areas or contexts where data are sparse.

Introduction

Carbonate reservoirs, despite their global significance in hydrocarbon production, present a persistent challenge to reservoir characterization due to their intrinsic heterogeneity. This complexity stems from a combination of depositional variability, diagenetic overprint, and the presence of diverse pore systems, which disrupt conventional petrophysical relationships. In this context, rock typing—the classification of reservoir rocks into groups with internally consistent flow and storage properties —emerges as a critical step in bridging geological description and dynamic reservoir modeling.

The necessity of rock typing in carbonate systems is not merely academic; it is vital. Unlike siliciclastic reservoirs, where depositional facies often correlate well with petrophysical behavior, carbonate rocks exhibit a wide spectrum of pore types and diagenetic alterations that hamper such correlations. Effective rock typing enables the grouping of rocks with similar petrophysical responses, thereby facilitating the assignment of porosity-permeability relationships, capillary pressure functions, and relative permeability curves within reservoir models (Lucia, 1995; Jennings & Lucia, 2003).

The central challenge involves transferring rock type information from core-scale observations to reservoir-scale models. Core data offer detailed insights into petrophysical properties, but their spatial coverage is limited, and extending these findings to three-dimensional reservoir grids can lead to significant bias and mistakes. Additionally, the spatial distribution of rock types is governed by geological processes that may not be well defined due to limited available data, particularly in areas or contexts where data are sparse.

This paper focuses on the development and application of genetic rock typing approaches, which aim to link geological processes—depositional and diagenetic—to petrophysical behavior. In particular, we explore the use of zero-dimensional (box) models that simulate the evolution of porosity and permeability through burial and fluid-rock interaction. These models are coupled with forward modeling frameworks that incorporate depositional facies and diagenetic histories, offering a predictive capacity for rock types even in intervals lacking core calibration. By integrating geological drivers into the rock typing workflow, this approach seeks to enhance the reliability of property distribution in reservoir models and improve the predictive power of simulation in carbonate reservoirs.

Extrapolating Rock Types

Challenges and methodological advances

The extrapolation of rock types from core-scale observations to full-field reservoir models remains one of the most persistent challenges in carbonate reservoir characterization. While rock typing provides a framework for assigning petrophysical properties, its utility depends critically on the ability to propagate these classifications into three-dimensional space. In carbonate systems, this task is complicated by abrupt lateral and vertical heterogeneity, diagenetic overprints, and the limited resolution of available data.

Traditional extrapolation methods often rely on facies models derived from core descriptions and well logs, assuming that depositional environments control petrophysical behavior. However, in carbonates, diagenesis frequently overrides depositional signatures, leading to poor correlation between facies and reservoir quality. Consequently, facies-based extrapolation tends to oversimplify the reservoir, masking critical heterogeneities that influence flow behavior.

Moreover, the resolution of well logs is insufficient to capture fine-scale features such as vugs or dissolution-enlarged pores, which may dominate flow in certain intervals. While logs provide continuous data, their inability to resolve secondary porosity or subtle diagenetic textures introduces uncertainty in rock type assignment. This is particularly problematic in data-sparse areas, where extrapolation must rely on analogs or stochastic methods, often with limited geological justification.

Among the approaches developed to address these limitations, the methodology proposed by [Rebelle & Lalanne \(2014\)](#) stands out for its clarity and operational relevance. His concept of Static Rock Types (SRT), developed within TotalEnergies, is based on the integration of core and log data across multiple scales to define geologically and petrophysically homogeneous rock groups. The SRT workflow reconciles plug-scale measurements with log-scale observations, ensuring that rock types are both meaningful and extrapolatable. Crucially, this approach is entirely data-driven, relying on observed petrophysical behavior and statistical clustering. As such, it cannot predict rock types that have not been directly observed, nor can it infer petrophysical properties in intervals lacking sufficient data coverage.

This limitation underscores a broader challenge in carbonate reservoir modeling: extrapolating geological drivers beyond well control. While geostatistics can reproduce spatial patterns using variograms or training images (Doligez et al., 2011), it often lacks geological grounding. Without clear geological rationale, stochastic realizations may mislead, offering precision without predictive value. In such cases, geostatistics risks becoming a space-filling tool rather than a modeling one. Nonetheless, progress is being made through logic-driven workflows and multi-scale data integration. Combining sedimentological interpretation, petrophysical clustering, and electrofacies modeling provides practical means to extend rock type definitions into uncored zones. These approaches do not resolve the fundamental uncertainty of extrapolation but offer a structured framework to manage it.

In conclusion, the extrapolation of rock types within carbonate reservoirs continues to be a significant and unresolved challenge. Although data-driven approaches like the SRT framework provide clear operational guidelines, their effectiveness is fundamentally constrained by the breadth of existing observations. The absence of predictive geological drivers in many carbonate systems continues to constrain the reliability of reservoir models. Addressing this challenge will require not only methodological innovation but also a deeper understanding of the geological processes that govern petrophysical variability at reservoir scale.

The Prevalent influence of Eogenesis on petrophysical heterogeneity in carbonate reservoirs

Carbonate reservoirs are renowned for their complex petrophysical heterogeneity, which arises from a succession of diagenetic processes occurring throughout their geological history. These processes are traditionally grouped into three stages: eogenesis (early diagenesis), mesogenesis (burial diagenesis), and telogenesis (uplift and near-surface alteration) (e.g., Choquette & Pray, 1970; Moore, 2001). While each stage contributes to the evolution of porosity and permeability, eogenesis plays a particularly decisive role in establishing the initial framework of heterogeneity that often persists through later modifications.

Eogenesis. Eogenesis occurs shortly after deposition, in surface environments influenced by marine or meteoric fluids. Because it closely follows sedimentation, early diagenetic processes are superimposed on active sedimentary fabrics, creating fine-scale, persistent heterogeneity. Microbial activity, early cementation, selective dissolution, and localized dolomitization dominate. Micritization alters grain textures and may reduce porosity. Marine cements like aragonite or high-Mg calcite form isopachous rims or botryoidal aggregates, either occluding pores or creating rigid frameworks. Early dolomitization, especially in tidal flats or lagoons, enhances porosity and permeability in a fabric-selective manner. Similarly, meteoric dissolution linked to freshwater lenses generates moldic or vuggy porosity by removing unstable minerals. These early alterations define the initial petrophysical framework, which influences subsequent diagenetic evolution and reservoir quality.

Early diagenetic features create small-scale heterogeneity that resists burial alteration. Cemented zones inhibit compaction and pressure solution, preserving adjacent porosity. Bioturbation and microbial coatings limit cementation, shaping pore architecture and fluid flow pathways at reservoir scale.

Mesogenesis. Mesogenesis occurs during deep burial under elevated temperature and pressure. It includes pressure solution, stylolitization, recrystallization, and cementation, which alter porosity and permeability without erasing early heterogeneity. Stylolites act as vertical barriers, guided by prior cemented zones. Burial dolomitization may enhance reservoir quality, but its effects remain superimposed on the pre-existing diagenetic fabric.

Telogenesis. Telogenesis involves uplift, exposure, and meteoric water influx, triggering karstification, oxidation, and fracturing. These processes enhance permeability, guided by earlier heterogeneities. Rigid, cemented or dolomitized zones fracture preferentially, channeling fluids that accelerate dissolution. Less altered areas undergo diffuse changes. Ultimately, telogenetic overprints reflect the initial petrophysical framework shaped during eogenesis and mesogenesis.

The dominant role of eogenesis in carbonate reservoirs is evident in case studies such as the Arab Formation in the Middle East. From a practical standpoint, recognizing the influence of eogenesis is essential for reservoir characterization and modeling. Early diagenetic features can often be identified in core and outcrop samples and sometimes inferred from well logs and seismic data. Incorporating eogenetic heterogeneity into static models improves the reliability of dynamic simulations, particularly in carbonate systems where early diagenesis governs the architecture of flow units. In conclusion, while mesogenesis and telogenesis contribute to the evolution of carbonate reservoirs, eogenesis is the stage that most strongly controls petrophysical heterogeneity. Its early imprint on sediment texture, mineralogy, and pore structure defines the pathways for later diagenetic processes and fluid flow. Understanding and modeling this early heterogeneity is crucial for accurate reservoir prediction, optimal well placement, and efficient hydrocarbon recovery.

A new methodology for Genetic Rock Typing in Carbonate Reservoirs

Conceptual foundation: A process-based framework for modeling and prediction

To address these constraints, the concept of genetic rock typing has been introduced as a comprehensive and predictive methodology. Adopting this approach enables us to systematically connect geological processes with reservoir performance, which is especially beneficial for extrapolation and forward modeling applications.

The foundations of genetic rock typing were laid by Lucia (1995, 1999, 2007), who emphasized the role of rock fabric and diagenetic history in controlling petrophysical properties. Rather than grouping rocks based solely on empirical porosity-permeability relationships, genetic rock typing seeks to classify them according to their evolutionary pathway—from deposition through diagenesis to burial. This approach recognizes that pore systems in carbonates are not static features but dynamic outcomes of geological processes.

At its core, genetic rock typing integrates three key controls: depositional fabric, which defines the initial pore architecture; early diagenetic modifications, such as cementation, micritization, or dissolution; and burial diagenesis, which includes compaction, recrystallization, and fracturing. By analyzing these stages in sequence, it becomes possible to reconstruct the genesis of pore systems and to classify rocks in a way that reflects their true reservoir potential. This process-based classification is particularly valuable in carbonate systems, where diagenetic overprints often obscure depositional signals and complicate petrophysical interpretation.

One of the principal advantages of genetic rock typing lies in its capacity for extrapolation. Because rock types are defined by geological processes rather than statistical patterns, they can be extended beyond cored intervals using concepts such as sequence stratigraphy, facies modeling, and diagenetic trends. For example, leached grainstones formed in high-energy shoal environments can be predicted to occur in specific stratigraphic positions, even in the absence of direct data. Similarly, the spatial distribution of early marine cements or meteoric dissolution features can be modeled based on paleogeographic and burial history, allowing for more accurate property assignment in reservoir grids.

This predictive capability is further enhanced when genetic rock types are coupled with forward modeling tools. Zero-dimensional (box) models, for instance, simulate the evolution of porosity and permeability through geological time, incorporating processes such as cementation, dissolution, and compaction. These models use depositional and diagenetic inputs to generate petrophysical envelopes that can be applied across the reservoir. When calibrated to regional geological histories, they provide a robust means of populating reservoir models with realistic property distributions.

In practice, genetic rock typing has proven its value in several carbonate reservoirs worldwide. In the Arab -D reservoir, genetic classifications helped differentiate high-permeability leached grainstones from low-permeability cemented packstones, improving history matching and waterflood performance. Similar

applications in the Khuff and Mishrif formations have demonstrated how genetic rock types can guide well placement, enhance recovery strategies, and reduce uncertainty in reservoir simulation.

Despite its strengths, genetic rock typing is not without challenges. It requires detailed geological knowledge and high-quality data, including petrographic analysis, core interpretation, and dynamic calibration. Moreover, the extrapolation of genetic rock types into uncored volumes must be done with care, as diagenetic processes can vary spatially and temporally. Nonetheless, when applied rigorously, this methodology offers a powerful tool for bridging geology and reservoir engineering.

In conclusion, genetic rock typing represents a paradigm shift in carbonate reservoir characterization. By grounding rock type definitions in geological processes, it enables more accurate extrapolation, more realistic modeling, and more predictive simulation. It is this approach that we adopt moving forward, confident in its ability to reconcile geological complexity with the demands of reservoir management.

Box Modeling in carbonate reservoirs

In carbonate reservoirs, where the prediction of petrophysical properties is complicated by the interplay of depositional architecture and diagenetic overprint, box modeling provides a pragmatic and efficient framework. This approach simplifies the reservoir into discrete volumetric units, or "boxes," each representing a distinct geological domain. The essence of box modeling is to explicitly link geological processes (e.g., sedimentation, diagenesis) to petrophysical properties, thereby simulating the possible evolution of rocks and their properties over time.

Our in-house box modeling application is founded on this philosophy. It uses box modeling as a strategic tool to integrate geological knowledge into reservoir models. By simulating the evolution of facies, it enables geoscientists to reconstruct the diagenetic history of a reservoir and validate it by calibrating against core data. This approach is particularly valuable as it provides reliable inputs for extrapolation and uncertainty quantification, even in undrilled areas.

The clarity and adaptability of box modeling make it ideal for rapid scenario testing and uncertainty analysis, which is of great value in large Middle Eastern carbonate fields. Rather than a mere simplification, this approach is a powerful tool for bridging the gap between geology and reservoir engineering. In this paper, we propose to develop and apply new box models adapted to carbonate systems, with an emphasis on extrapolation, forward modeling, and uncertainty quantification.

Methodological Framework for Predictive Rock Typing

A coupling between box-modeling and forward modeling. Predicting petrophysical properties in carbonate reservoirs requires a methodological framework capable of integrating both geological processes and their dynamic evolution. To address this challenge, we propose a coupled approach, combining two complementary in-house tools dedicated to carbonates.

Our box modeling application predicts petrophysical properties by superimposing diagenetic transformations onto an initial sedimentary architecture. It generates "rock types" that reflect both their sedimentary origin and the diagenetic alterations that have affected them.

In parallel, a process-based forward modeling software has been developed. It reconstructs the stratigraphic architecture and the reservoir step-by-step and in 3D, by reproducing sedimentary processes (transport, currents, erosion, basement deformation) and diagenetic processes operating in different hydrozones (e.g., marine, phreatic, vadose) at each time step. It is a truly dynamic modeling tool that simulates the reservoir's development over time (Danquigny et al., 2025).

The key innovation of our approach lies in the direct coupling between box modeling and forward modeling. The box-model serves not only as a model generator but also as a validation and calibration framework for the results from the forward modeling. This dual functionality enables the iterative refinement of model parameters and ensures that the results from the complete forward modeling are perfectly

consistent with the observed "rock types." A crucial development of this coupling is that both tools utilize identical mineral reaction schemes, which ensures an inherent consistency in the diagenetic simulation. In essence, this integrated methodology represents a significant advance in modeling, as it reconciles static and dynamic approaches for a more reliable prediction and extrapolation of rock types.

Hydrozones, diagenetic zone, and diagenetic reactions. The predictive power of our coupling relies on a robust geological foundation. The spatial organization of diagenetic reactions in our model is based on a structured zonation of the subsurface, built around the concept of hydrozones. These are hydrogeological domains characterized by distinct water flow regimes and geochemical conditions. Each hydrozone is further subdivided into diagenetic zones where specific reactions—mineral stabilization, dissolution, and cementation—are dominant. These reactions are modeled to reflect the evolution of porosity types (interparticle, moldic, vuggy) and mineralogical composition, thereby influencing the overall reservoir quality. The model identifies four main hydrozones, each with multiple diagenetic zones. The table below shows how the hydrozones, diagenetic zones, and modeled reactions correspond in both softwares.

Within the meteoric vadose zone, located above the Free Water Level (FWL), the upper vadose zone is dominated by mineral stabilization and dissolution. As meteoric water percolates downwards, it becomes supersaturated with respect to calcite due to this dissolution. This leads to a shift toward cementation in the lower vadose zone, where the precipitation of meniscus cements fills pore spaces, indicating increased water-rock interaction and the onset of mineral precipitation.

The meteoric phreatic zone, fully saturated with freshwater between the FWL and the Base of Free Water Level (BFWL), includes the water table fringe, where all three reactions—stabilization, dissolution, and cementation—are active. This zone represents a dynamic interface where chemical gradients drive complex diagenetic transformations. Below it, the freshwater lens is characterized primarily by stabilization, suggesting a more chemically stable environment with limited aggressive dissolution or cementation.

The transition zone, located between the BFWL and the Base of Mixing (BM), includes two distinct environments. The mixing zone is chemically reactive, with both stabilization and dissolution occurring due to the interaction of meteoric and marine waters. The intertidal zone, a shallow layer at the interface between sea and rock, is dominated by cementation, driven by marine water saturation and tidal pumping.

Table 1—Relationship between modeled diagenetic reactions, diagenetic zones, and hydrozones where the reactions take place

Hydrozone	Diagenetic zone	Diagenetic reaction		
		Mineral stabilization	Dissolution	Cementation
Meteoric vadose	Upper vadose	✓	✓	
	Lower vadose	✓		✓
Meteoric phreatic	Water table fringe	✓	✓	✓
	Freshwater lens	✓		
Transition	Intertidal			✓
	Mixing	✓	✓	
Marine phreatic	Sea floor			✓
	Shallow burial		✓	
	Marine burial	✓		✓

In the marine phreatic zone, which lies below the BM and is saturated with seawater, three diagenetic zones are defined. The sea floor is primarily affected by cementation, reflecting active marine precipitation. The shallow burial zone is dominated by dissolution, likely due to organic matter degradation and fluid

migration. The marine burial zone combines stabilization and cementation, indicating deeper, more stable conditions with ongoing mineral precipitation.

In addition to chemical reactions, the model incorporates compaction, both syndimentary and burial, as a mechanical process that reduces sediment thickness and porosity over time. Compaction is simulated based on sediment accumulation rates and rock fabric, ensuring that mechanical diagenesis is integrated alongside chemical processes.

This zonation and reaction framework provides a robust basis for predictive modeling in carbonate reservoirs. By linking hydrogeological context to diagenetic evolution, it enables the simulation of petrophysical properties in a geologically consistent manner, supporting both static characterization and dynamic reservoir forecasting.

Application to the Urgonian Formation of the ALBION Project (SE France)

Methodology

To illustrate and validate our approach, we apply it to the Urgonian Formation, a Late Barremian-Early Aptian unit from the Southeast France Basin. Interestingly, these Urgonian limestones are outcrop analogues of Middle East carbonate reservoirs (Thamama, Kharaib and Shuaiba formations; [Alsharhan and Nairn, 1997](#)) from different points of view: age, facies, paleoenvironments and reservoir properties (pore types, porosity, permeability; cf. [Borgomano et al., 2002](#); [Sallier and Hamon, 2005](#); [Leonide et al., 2012](#)). The architecture of the Urgonian carbonates, and consequently the distribution of reservoir properties, is the result of a complex interplay between accommodation rate, eustasy, and tectonics. Regional studies have shown that this platform is a highly heterogeneous mosaic of microporous and heavily cemented carbonates, making it an ideal case study for our method ([Leonide et al., 2014](#); [Tendil et al., 2018](#); [Barbier et al., 2025](#)).

Instead of relying on a petrographic analysis of core samples, our study is based on the results of forward stratigraphic modeling. These models were calibrated using reference data (cores, logs, geological data) from studies on the Provence Urgonian. From these digital models, we have extracted virtual samples that correspond to various sedimentary facies and have undergone a complex diagenetic history, allowing us to study their petrophysical evolution.

Geological settings

The evolution of the Urgonian Formation is characterized by significant geodynamic influences that have shaped both its structural architecture and subsequent diagenetic processes. The carbonate platform in Southeast France developed within a divergent tectonic regime yet was substantially impacted by regional and global tectonic as well as eustatic events ([Tendil et al., 2018](#); [Barbier et al., 2025](#)). Through forward modeling, we are able to simulate and systematically analyze this geological history and its effects on diagenesis and petrophysical properties:

Depositional Period. During the Barremian-Aptian, sedimentation was primarily controlled by tectonic, climate, and eustatic fluctuations. The platform developed through progradation and aggradation cycles, which created a succession of facies (inner platform, shoals, pelagic facies) and their initial pore architecture. Syn-sedimentary discontinuities linked to periods of subaerial exposure were identified. These periods significantly affected the pore networks of the Urgonian carbonates ([Leonide et al., 2014](#)).

Uplift (Cenozoic). After deposition, the platform was sub-aerially exposed and subjected to early diagenetic processes. Later, during the Cenozoic, the Durance Uplift, linked to the tectonic movements of the Alps, caused significant uplift and erosion of the northern part of the reservoir. This uplift exposed the carbonates to meteoric waters, involving intense microporosity development and mineral stabilization.

Burial. The period between the Durancien uplift and the current state was dominated by progressive burial and associated calcite cementation.

Current State. The geometry and reservoir properties we observe today are the cumulative result of these processes. Faults and fractures that formed during burial and tectonic phases acted as conduits for meteoric waters, creating dissolution pathways and large-scale heterogeneities.

Depositional Facies Associations and their sedimentary context

The foundation of any rock typing workflow lies in the accurate identification and classification of depositional facies. In our approach, these facies associations (FA) are defined based on their sedimentary environment, texture, fossil content, and mineralogical composition. Each facies serves as the starting point for diagenetic modeling and ultimately determines the petrophysical trajectory of the resulting rock types.

In our study, the forward stratigraphic models have allowed us to generate geologically plausible scenarios and to extract model cells (i.e., virtual samples) for which we wanted to verify the diagenetic pathways and the evolution of petrophysical properties with the boxmodeling. Utilising these "samples," we have identified the principal facies associations (see Fig. 1) as follows:

FA2: Inner Platform Facies. This association is found in low-energy, shallow environments, dominated by mudstones and wackestones. It also includes more energetic parts of the inner platform, which explains the presence of peloidal and foraminifera packstones that can sometimes become grainstones. These facies are characterized by the presence of miliolids, gastropods, stromatolites, and coprolites. These facies are often associated with decametric cyclicities and are the site of intense early diagenesis (Leonide et al., 2014).

FA3: Inner Platform Rudist Facies. This association is located in the deeper parts of the inner platform and is dominated by rudist-rich facies, forming "patches" several kilometers wide. These patches, which can be described as "bioherms," are typically found at a depth of a few meters to several tens of meters.

FA5: Coral Facies. This association is composed of coral wackestones to grainstones and also forms patches of several hundred meters to several kilometers in width. It is in lateral transition from the rudist facies (FA3) and represents bio-constructed barriers that protect the inner platform.

FA6 & FA7: Shoal Facies. These associations represent high-energy environments on the outer platform. FA6 is characterized by oobioclastic shoals, while FA7 is composed of bryozoan and echinoderm sands, typical of deeper bathymetries. These facies are often associated with specific sedimentary structures (foresets, clinobeds) that influence the distribution of petrophysical properties (Leonide et al., 2012).

FA11: Pelagic Facies. This is an association of fine-grained micritic limestone found in deep hemipelagic and pelagic environments. This facies is characterized by sponge spicules and shells of marine organisms such as ammonites and belemnites. Slumping features and hardgrounds are commonly observed, which indicates discontinuous sedimentation and slope instability (Tendil et al., 2018). Each of these facies associations provides a distinct starting point for diagenetic evolution. Their mineralogical composition, sedimentary texture, and biological content influence the pathways through which they transform into specific rock types. In box-models, these facies are not static descriptors but dynamic inputs into a process-based model that simulates their transformation under varying diagenetic conditions.

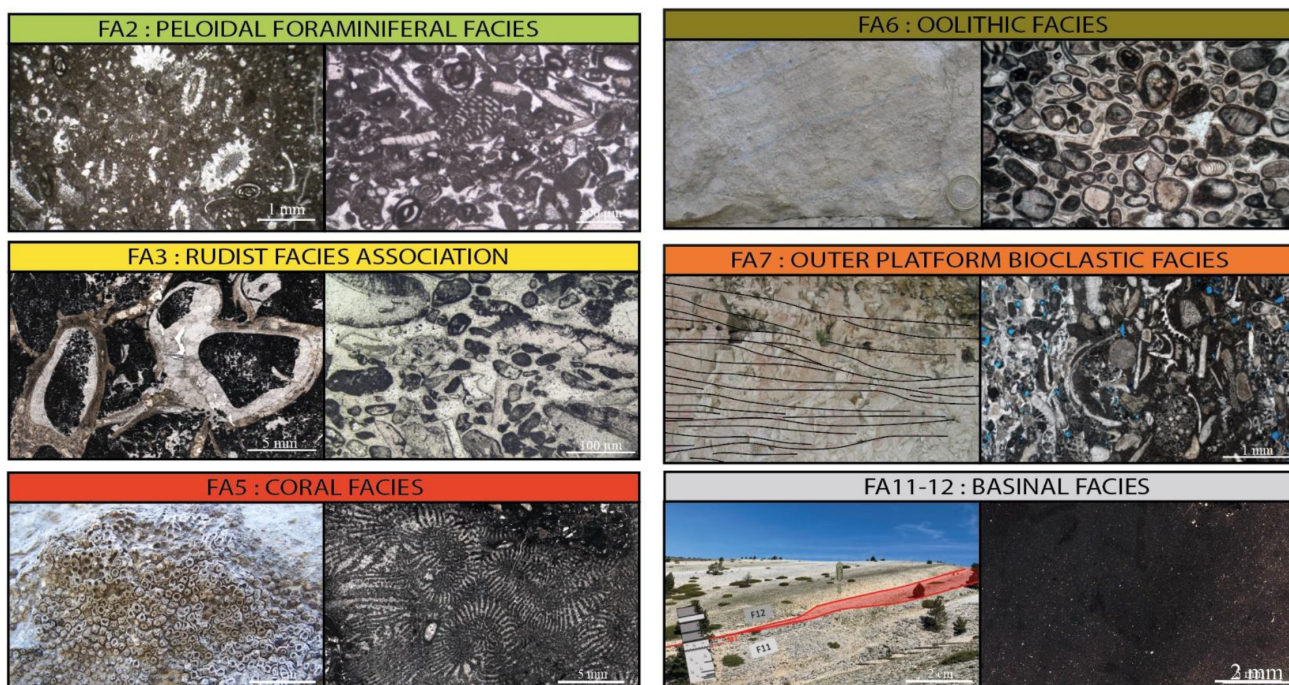


Figure 1—Microscopic and macroscopic illustration of the main depositional facies present in the reservoir of this case study.

Rock type classification based on petrophysical properties

Unlike traditional classifications based solely on depositional facies, our "genetic rock typing" method classifies rock types (RT) based on petrophysical signatures. This approach is particularly critical because distinct rock types, with different diagenetic histories, may appear similar on a porosity-permeability cross-plot. Each rock type is defined by a specific porosity and permeability range that reflects the cumulative impact of initial mineralogical composition, sedimentary texture, and diagenetic history. Each rock type is defined by a specific porosity and permeability range that reflects the cumulative impact of initial mineralogical composition, sedimentary texture, and diagenetic history. Figure 2 illustrates the contrast between a facies-based classification and a rock type-based classification, highlighting the added value of our approach.

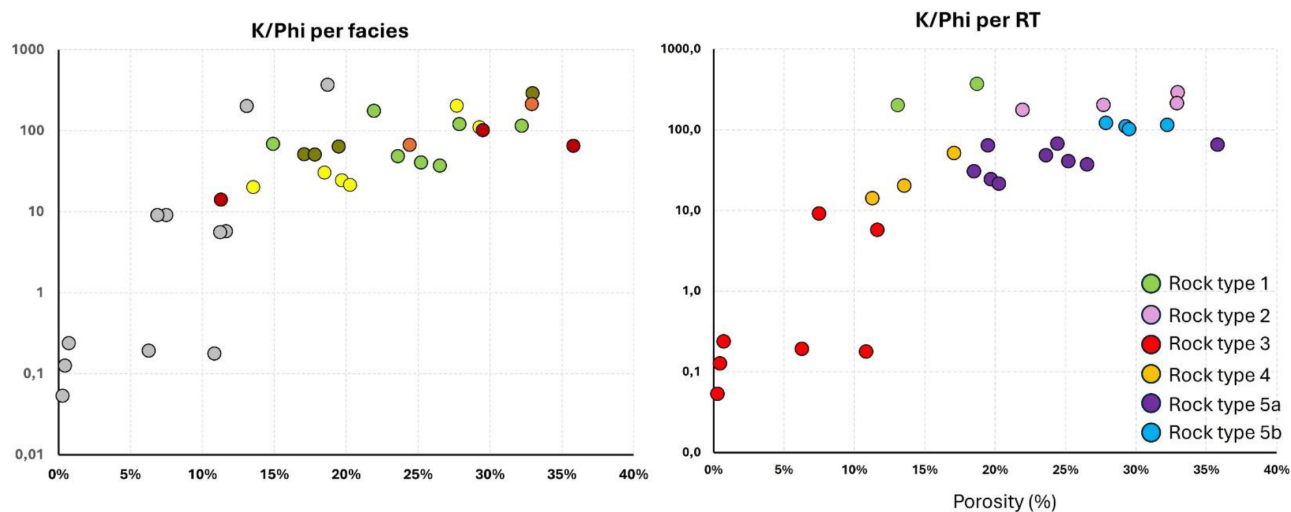


Figure 2—Porosity & Permeability colored by facies (left) or by rock-type (right).

Following analysis of cells obtained from our stratigraphic models, we have identified six primary rock types as outlined below:

Rock Type 1 (RT1). It has a moderate porosity, mainly between 12% and 18%. Its permeability is very high, with values ranging from 200 mD to over 300 mD.

Rock Type 2 (RT2). It is characterized by high porosity, in a wide range of 20% to 35%, and high permeability, from 150 mD to over 292 mD.

Rock Type 3 (RT3). This rock type has very low porosity, generally below 15%, and extremely low permeability, almost always below 10 mD.

Rock Type 4 (RT4). It has low to moderate porosity, ranging from 10% to 20%, and moderate permeability, between 10 mD and 70 mD.

Rock Type 5a (RT5a). Its porosity falls within a range of 18% to 27%, with permeability varying from 20 mD to 98 mD.

Rock Type 5b (RT5b). It has high porosity, between 25% and 33%, and high permeability, in the range of 100 mD to 200 mD.

This classification not only organizes data but also creates a direct correlation between petrophysical properties and the simulated geological history. This facilitates the shift from a descriptive framework to a predictive modeling approach for the reservoir.

Diagenetic Pathways and Rock Type Evolution

In our box-model, these diagenetic pathways are modeled as sequences of residence within specific hydrozones and diagenetic zones, each associated with characteristic reactions—cementation, dissolution, and mineral stabilization. The cumulative effect of these reactions determines the final petrophysical properties of the rock.

Each rock type is associated with one or more diagenetic pathways, which may diverge or converge depending on the facies origin and the sequence of diagenetic environments encountered. For example, RT1 and RT3 both originate from FA11, a pelagic mudstone rich in high-magnesium calcite (HMC). However, RT1 follows a pathway involving early cementation on the seafloor, dissolution in shallow marine burial, prolonged stabilization in the freshwater lens, and final dissolution in the upper vadose zone during Durancian uplift. This sequence promotes the development of a vuggy and moldic pore network, resulting in high permeability. In contrast, RT3 undergoes similar early stages but ends with late cementation in the lower vadose zone during the same tectonic episode in Cenomanian, effectively occluding the pore system and producing low permeability.

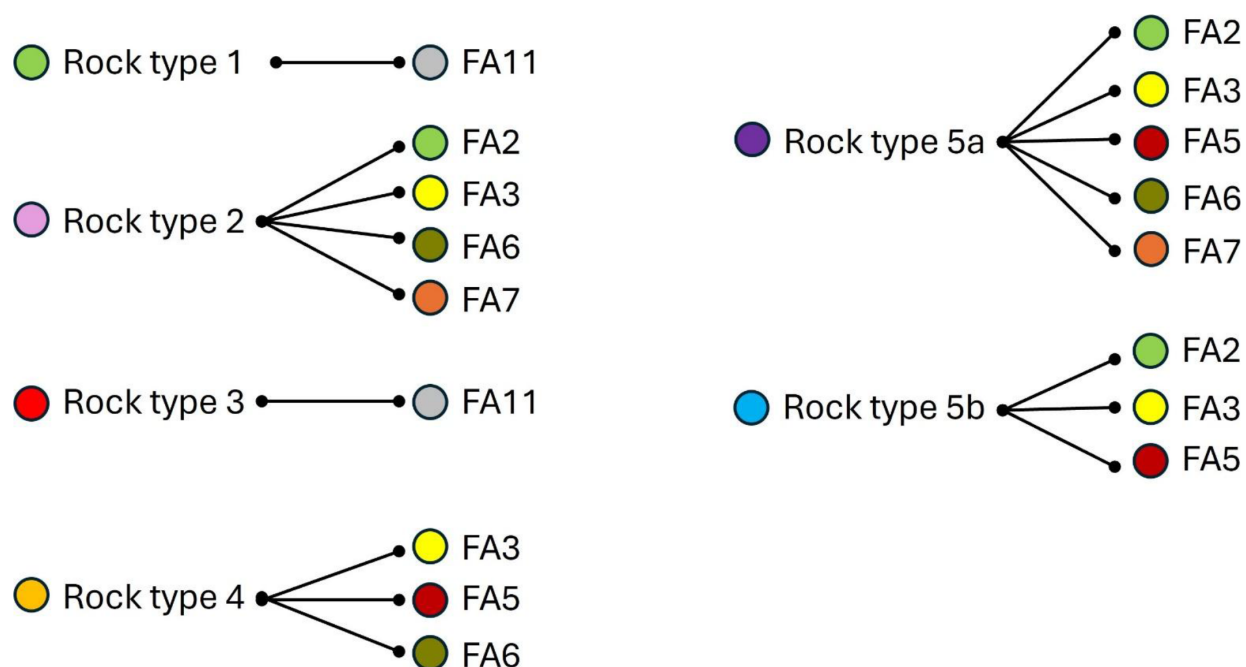


Figure 3—Correspondence between Facies Association and Rock-Types. For example, the FA11 facies can become two different rock types: rock type 1 and rock type 3. All other facies, depending on their diagenetic pathway, can become rock type 2, 4, 5a, and 5b.

RT2 illustrates a case of diagenetic convergence, where different facies —FA3 (rudist packstone) and FA6 (ooid grainstone)—follow comparable diagenetic sequences and arrive at similar petrophysical endpoints. Both experience initial seafloor cementation, repeated early transitions through mixing zones with alternating dissolution and stabilization during Barremian sedimentation, and final residence in the freshwater lens during the Durancian uplift. Despite their distinct sedimentary origins, the shared diagenetic history leads to comparable porosity and permeability values.

RT4, RT5a and RT5b further demonstrate the influence of diagenetic timing and residence duration. RT4 results from granular facies (FA2, FA3, FA6) undergoing alternating marine burial and meteoric diagenesis, with a final cementation phase in the lower vadose zone during the Durancian uplift. RT5a and RT5b share a common early history but diverge in their final stages: RT5b remains longer in the freshwater lens during the Durancian uplift, preserving porosity, while RT5a transitions into the lower vadose zone, where cementation reduces reservoir quality.

Our software tracks these pathways using residence time data and reaction sequences, allowing for the reconstruction of petrophysical evolution. The model captures both the divergence of rock types from single facies and the convergence of multiple facies toward a common rock type, emphasizing the central role of diagenesis in reservoir heterogeneity.

Description of the diagenetic pathways

Towards RT1. The formation of RT1 is directly linked to the exhumation and dissolution phases associated with the Durance Uplift. The diagenetic pathway begins with early cementation on the seafloor. The samples then undergo a progressive burial that brings them to the shallow marine phreatic zone, where a phase of dissolution of metastable minerals occurs. It is during the Durance Uplift that the sample is brought back towards the surface, which causes it to undergo prolonged stabilization in the freshwater lens (for nearly 10 to 20 million years). An intense dissolution in the upper vadose zone completes the formation of a vuggy and moldic pore network, explaining its high permeability.

Towards RT2. The pathway for RT2 is often linked to early subaerial exposure events that occurred during the depositional period of the Urgonian platform. Samples from these facies may first undergo cementation on the seafloor, then experience very early dissolution and stabilization phases due to these local or global synsedimentary exposures. These samples will then pass into the shallow burial zone of the marine environment, where dissolution occurs. Then, in the Cenomanian, they undergo the mixing zone and freshwater lens processes, which lead to their final high porosity and permeability values. This shared diagenetic history, rather than their distinct sedimentary origins, is what explains the comparable petrophysical properties of RT2.

Towards RT3. The defined rock type 3 is initially composed entirely of FA11 facies. These, like rock type 1, potentially undergo cementation on the seafloor before experiencing phases of dissolution and cementation, respectively, during progressive burial in the shallow marine burial and marine burial diagenetic zones. It is then during the Durance Uplift that these samples reach the freshwater lens, where a phase of mineralogical stabilization takes place. However, unlike RT1, RT3 undergoes a late cementation phase in the lower vadose zone. This cementation is the key event that occludes the pore system and explains its extremely low permeability.

Towards RT4. The formation of rock type 4 results from two distinct diagenetic pathways, both beginning with the alteration of granular facies originating from Facies Associations FA2, FA3, and FA6. The primary pathway involves these facies following a sequence of processes identical to those defining rock type 3, by undergoing diagenesis in the shallow marine burial zone and potentially the marine burial zone. An alternative pathway sees these same facies undergo repeated transitions between the shallow marine burial zone and a freshwater lens during sedimentation, a process linked to successive subaerial exposure events. Regardless of their initial path, the two pathways converge toward the same final phases: a mandatory passage through a freshwater lens and into the lower vadose zone in Cenomanian during Durancian uplift, where a terminal cementation phase occurs that completes their formation.

Towards RT5. The two subgroups of Rock Type 5, Rock Type 5A and Rock Type 5B, share a similar overall diagenetic history, yet exhibit a significant difference in porosity and permeability, with Rock Type 5B being consistently higher. The common diagenetic pathway for both begins with a potential early cementation on the seafloor. Subsequently, the samples undergo successive stages in the marine realm, moving through the shallow burial zone and then the deeper burial zone, before returning to a meteoric influence during Durancian uplift. The crucial distinction in their final diagenetic history accounts for their different petrophysical properties. Rock Type 5B spends more time in the freshwater lens at the end of its diagenesis and remains there, promoting greater preservation of porosity. In contrast, Rock Type 5A passes from the freshwater lens into the lower vadose zone, where it undergoes a terminal cementation phase. This late cementation, combined with the fact that Rock Type 5A generally spends more time in the marine burial zone where a cementation phase occurs, leads to its lower porosity and permeability compared to Rock Type 5B.

Facies-Rock Type: mapping and prediction

A central feature of the box-modeling methodology is its ability to translate depositional facies into petrophysically meaningful rock types through process-based diagenetic modeling. This transformation is not deterministic; rather, it reflects the diversity of diagenetic pathways that a given facies may follow, depending on its residence time, sequence of reactions, and hydrogeological context.

The mapping between facies and rock types reveals both divergent and convergent trajectories. Divergence occurs when a single facies, such as FA11, evolves into multiple rock types depending on its diagenetic history. For instance, FA11 can become RT1 if it undergoes prolonged dissolution and stabilization in meteoric zones, or RT3 if it experiences early cementation in the lower vadose zone (Fig.4).

Despite identical sedimentary origins, the resulting petrophysical properties differ dramatically, with RT1 exhibiting high permeability and RT3 showing near-complete pore occlusion.

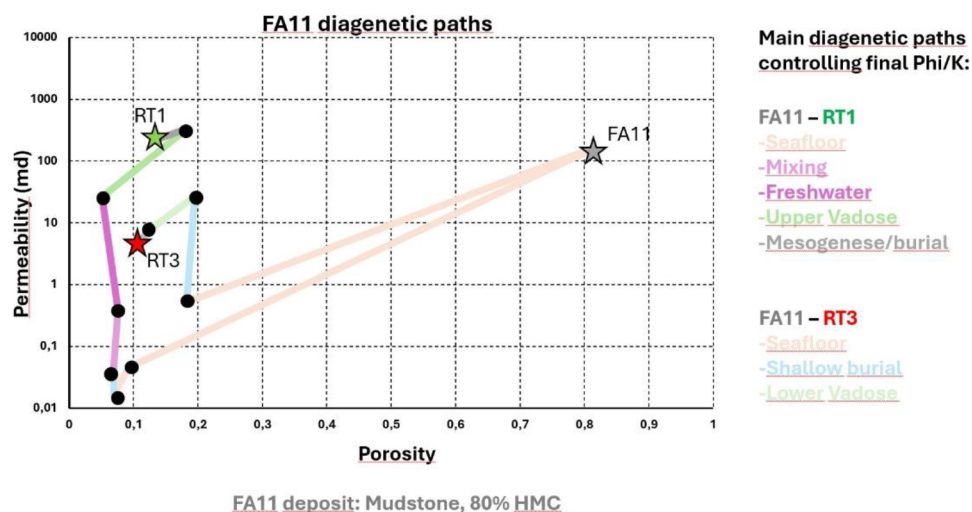


Figure 4—Divergence example of FA11 becoming RT1 and RT3.

Conversely, convergence is observed when distinct facies—such as FA3 (rudist packstone) and FA6 (ooid grainstone)—follow similar diagenetic sequences and arrive at the same rock type, RT2 (Fig.5). In this case, both facies undergo initial seafloor cementation, repeated transitions through mixing zones, and final residence in the freshwater lens. The shared diagenetic logic overrides their sedimentary differences, resulting in comparable porosity and permeability.

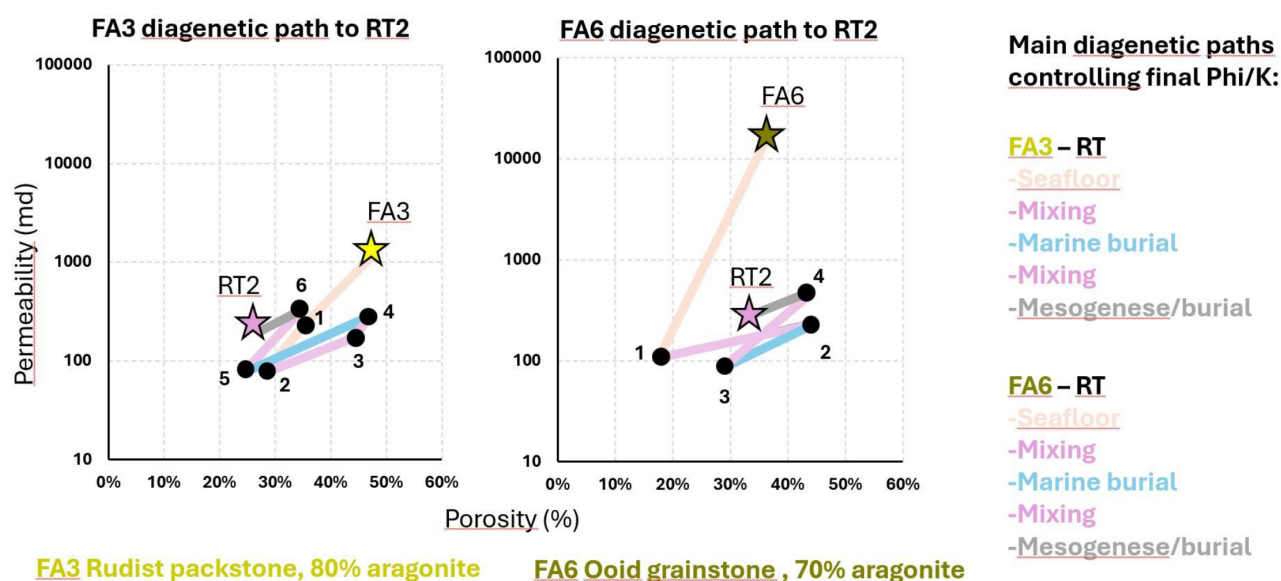


Figure 5—Convergence example of two sedimentary facies FA3 and FA6 becoming RT2.

In box-modeling, facies-rock type logic is formalized as a matrix of possible transformations, each controlled by a specific diagenetic history. The model accounts for the timing and intensity of reactions, the mineralogical composition of the facies, and the sequence of hydrozones encountered. By simulating these interactions, the software generates predictive maps that link facies distributions to expected rock types, enabling geoscientists to anticipate reservoir quality in uncored or poorly sampled areas.

Case Study conclusions

Using virtual samples extracted from forward stratigraphic models of the Urganian Formation of the ALBION case, the methodology identifies key facies associations and simulates their diagenetic evolution through sequences of cementation, dissolution, and mineral stabilization. The resulting rock types are classified based on porosity and permeability, revealing both divergent and convergent pathways depending on facies origin and diagenetic history.

The genetic rock typing approach adopted here moves beyond traditional facies-based classification by defining rock types according to their petrophysical signatures—specifically porosity and permeability ranges. Six primary rock types are identified, each reflecting a unique diagenetic history. For instance, RT1 and RT3 both originate from pelagic facies (FA11) but diverge due to differences in late-stage cementation, resulting in contrasting permeability. Similarly, RT2 exemplifies diagenetic convergence, as rudist packstones (FA3) and ooid grainstones (FA6) follow similar reaction sequences and yield comparable petrophysical properties.

The model captures both divergence and convergence in facies-to-rock type evolution, emphasizing the central role of diagenesis in reservoir heterogeneity. RT4, RT5a and RT5b further illustrate how timing and residence duration in meteoric zones influence final reservoir quality. RT5b, for example, retains higher porosity due to prolonged stabilization in the freshwater lens, whereas RT5a undergoes terminal cementation in the lower vadose zone, reducing its permeability.

Conclusions

The predictive modeling of rock types in carbonate reservoirs demands more than empirical classification—it requires a geologically grounded understanding of how depositional and diagenetic processes interact over time. Genetic rock typing addresses this need by linking depositional fabric, early diagenesis, and burial history to pore system evolution. This approach transforms petrophysical classification into a causal and predictive framework, particularly valuable in data-sparse environments.

To address the persistent challenge of rock type prediction and extrapolation in carbonate reservoirs, this study proposes a unified framework that integrates box modeling with forward modeling. These two approaches are not only complementary but also deeply interconnected, offering a process-driven methodology grounded in geological logic. Box modeling simulates the transformation of depositional facies into rock types through sequences of diagenetic reactions—cementation, dissolution, and mineral stabilization—based on their residence in specific hydrozones and diagenetic zones. Each facies is assigned to a box defined by its depositional context and mineralogical composition, resulting in petrophysical predictions that reflect both origin and diagenetic trajectory.

The integration of box modeling into forward reservoir modeling workflows represents a methodological advance. A critical requirement for this integration is the consistency of mineral reaction schemes across both tools. Defining cementation, dissolution, and stabilization identically ensures coherent simulation of diagenetic evolution and preserves geological validity.

This coupling enables iterative calibration between modeled sedimentary architectures and petrophysical outcomes. By comparing the outputs of forward models with the rock types predicted by box modeling, geoscientists can adjust parameters to ensure consistency and reliability. This feedback loop strengthens the extrapolation of reservoir properties into uncored volumes and enhances confidence in predictive modeling.

Together, box modeling and forward modeling provide a geologically faithful framework for resolving the long-standing issue of rock type prediction. Box modeling offers a predictive engine for rock type evolution based on diagenetic logic, while forward modeling simulates the dynamic interplay of sedimentation, early diagenesis, and compaction in three dimensions. Their integration supports a more accurate and traceable workflow, bridging the gap between geological understanding and reservoir

simulation. This approach lays the foundation for improved reservoir characterization and modeling fidelity in complex carbonate systems.

The application of this methodology to the Provence Urgonian platform has validated its robustness. The integration of data from this study successfully demonstrated how the approach captures both diagenetic divergence and convergence, highlighting the central role of diagenesis in creating reservoir heterogeneity.

Ultimately, genetic rock typing bridges the gap between geological understanding and reservoir simulation. It offers a robust, process-oriented foundation for more accurate forward modeling and more reliable prediction of reservoir behavior. As modeling workflows continue to evolve toward greater geological fidelity, this methodology is poised to play a central role in the future of carbonate reservoir modeling.

Acknowledgement

The authors gratefully acknowledge TotalEnergies for their support in funding the research and development of CRONOS™, the proprietary geological forward reservoir simulation suite designed for carbonate reservoirs. Within this suite, SED-RES™ serves as the dedicated tool for sedimentary forward modeling, encompassing depositional processes, diagenesis, compaction, and petrophysical prediction. Additionally, PETROCARB™ is an in-house box modeling tool that simulates petrophysical evolution over time. SED-RES™ and PETROCARB™ were both essential components in the research conducted for this study.

The authors would also like to express their sincere gratitude to TotalEnergies for granting permission to publish this work.

Furthermore, sincere thanks are given to all collaborators of the CRONOS™ team for their essential assistance throughout the development and evaluation phases of the diagenetic and petrophysical modules of both SED-RES™ and PETROCARB™. In particular, the authors wish to recognize Cedric Gal, Emmanuelle Leonforte, Pascal Siegel, Deborah Siffert, and Jeremy Ruiiu for their valuable collaboration and commitment.

References

- Alsharhan, A.S., Nairn, A.E.M., 1997. *Sedimentary Basins and Petroleum Geology of the Middle East*. Elsevier, Amsterdam.
- Barbier M., Borgomano J., Leonide P., Massonnat G., Danquigny C., Lesueur J.L., 2025. Quantitative sequence stratigraphy applied to the Barremian-lower Aptian Urgonian carbonates deposited in Provence (southeastern France). *AAPG Bulletin*, v. **109**, no. 4, pp. 467–496.
- Borgomano, J., Masse, J.P., Al Maskiry, S., 2002. The Early Aptian Shuaiba carbonate outcrops in Jebel Akhdar, northern Oman: Impact on static modeling for Shuaiba petroleum reservoirs. *AAPG Bulletin* **86**, 1513–1529. doi: [10.1306/61EEDCE2-173E-11D7-8645000102C1865D](https://doi.org/10.1306/61EEDCE2-173E-11D7-8645000102C1865D).
- Choquette, P.W., Pray, L., 1970. Geologic nomenclature and classification of porosity in sedimentary carbonates. *AAPG Bull.*, **54**, 207250.
- Danquigny C., Barbier M., Dal Soglio L., Labeur A., Michel J., Lesueur J.L., and Massonnat G.J., 2025, Predicting the Distribution of Petrophysical Properties in Carbonate Reservoirs Using Process-based Forward Modeling, ADIPEC conference, Abu Dhabi, SPE 229530.
- Doligez B., Hamon Y., Barbier M., Nader F., Lerat O., Beucher H., 2011. Advanced workflows for joint modelling of sedimentary facies and diagenetic overprint—Impact on reservoir quality. SPE 146621, SPE ATCE, Denver, USA.
- Jennings, J.W., Lucia F.J., 2003. Predicting Permeability From Well Logs in Carbonates With a Link to Geology for Interwell Permeability Mapping. *SPE Reservoir Evaluation & Engineering*, **6**(4):215–225. doi: [10.2118/84942-PA](https://doi.org/10.2118/84942-PA).
- Leonide P., Borgomano J., Masse JP, Doublet S. 2012. Relation between stratigraphic architecture and multi-scale heterogeneities in carbonate platforms: The Barremian-lower Aptian of the Monts de Vaucluse, SE France. *Sedimentary Geology* **265**: 87–109. doi: [10.1016/j.sedgeo.2012.03.019](https://doi.org/10.1016/j.sedgeo.2012.03.019).
- Leonide P., Fournier F., Reijmer J.J.G., Vonhof H., Borgomano J., Dijk J., Rosenthal M., Van Goethem M., Cochard J., Meulenaars K., 2014. Diagenetic patterns and pore space distribution along a platform to outer-shelf transect (Urgonian limestone, Barremian- Aptian, SE France). *Sedimentary Geology* **306**, 1–23.

- Lucia, F.J., 1995. Rock-fabric/petrophysical classification of carbonate pore space for reservoir characterization. *AAPG Bulletin*, **79**, 1275–1300.
- Lucia, F.J., 2007. Carbonate Reservoir Characterization: An Integrated Approach. 2nd Edition, Springer-Verlag, Heidelberg.
- Machel, H.G., 2004. Concepts and models of dolomitization: a critical reappraisal, in Braithwaite, C.J.R., Rizzi, G., and Darke, G. eds., *The Geometry and Petrogenesis of Dolomite Hydrocarbon Reservoirs: Geological Society*, London, Special Publication, No. 235, p. 7–63.
- Moore, C.H., 2001. Carbonate Reservoirs - Porosity Evolution and Diagenesis in a Sequence Stratigraphic Framework. Elsevier Science, Developments in Sedimentology, Tome 55, 460 p.
- Rebelle M., Lalanne B., 2014. Rock-typing In Carbonates: A Critical Review Of Clustering Methods. ADIPEC conference, Abu Dhabi, SPE 171759.
- Sallier, B., Hamon, G., 2005. Micritic limestones of the Middle East: influence of wettability, pore network and experimental technique on drainage capillary pressure curve. *Int. Symp. of the Society of Core Analysts*, SCA2005-08, pp. 1–14.
- Tendil AJ-B, Frau C, Léonide P, Fournier F, Borgomano JR, Lanteaume C, Masse J-P, Massonnat G, Rolando J-P. 2018. Platform-to- basin anatomy of a Barremian-Aptian Tethyan carbonate system: New insights into the regional to global factors controlling the stratigraphic architecture of the Urgonian Provence platform (southeast France). *Cretaceous Research* **91**: 382–411. doi: [10.1016/j.cretres.2018.05.002](https://doi.org/10.1016/j.cretres.2018.05.002).